Definitions: Carré du Champ and Normalised Diffusion

- **Carré du Champ Operator:** For an infinitesimal generator $\mathcal{L}$ of a Markov diffusion process, the bilinear form is defined as:

$$\Gamma_{\mathcal{L}}(f, h) := \frac{1}{2}\big(f\mathcal{L}h + h\mathcal{L}f - \mathcal{L}(fh)\big) \tag{3}$$

- **Normalised Diffusion Operator:** Approximates the geometry from discrete samples $X \sim p$ using a Gaussian heat kernel $w_\epsilon(\boldsymbol{y}, \boldsymbol{x}) = \exp\left(-\frac{\|\boldsymbol{y}-\boldsymbol{x}\|^2}{2\epsilon}\right)$:

$$(P_\epsilon f)(\boldsymbol{y}) := \frac{(\mathcal{K}_\epsilon f)(\boldsymbol{y})}{d_\epsilon(\boldsymbol{y})}$$

- **Kernel Operator and Degree Function:**

$$(\mathcal{K}_\epsilon f)(\boldsymbol{y}) := \mathbb{E}_{X\sim p(x)}[w_\epsilon(\boldsymbol{y}, X) f(X)]$$

$$d_\epsilon(\boldsymbol{y}) := \mathbb{E}_{X\sim p(x)}[w_\epsilon(\boldsymbol{y}, X)]$$

Derivation: Substituting $\mathcal{L}_\epsilon$ into the CDC Operator (Part 1)

- **Initial CDC Definition:**
$$\Gamma_{\mathcal{L}_\epsilon}(f,h) = \frac{1}{2}(f\mathcal{L}_\epsilon h + h\mathcal{L}_\epsilon f - \mathcal{L}_\epsilon(fh))$$
- **Substitute $\mathcal{L}_\epsilon = \frac{1}{\epsilon}(\mathbf{I} - P_\epsilon)$ evaluated at y:**
$$\begin{aligned}\Gamma_\epsilon(f,h)(y) = \frac{1}{2\epsilon}\Big[& f(y)\big(h(y) - P_\epsilon h(y)\big) \\ & + h(y)\big(f(y) - P_\epsilon f(y)\big) \\ & - \big(f(y)h(y) - P_\epsilon(fh)(y)\big)\Big]\end{aligned}$$
- **Expand and cancel scalars** $(+f(y)h(y)$ and $-f(y)h(y))$:
$$\begin{aligned}\Gamma_\epsilon(f,h)(y) = \frac{1}{2\epsilon}\Big[& f(y)h(y) - f(y)P_\epsilon h(y) \\ & - h(y)P_\epsilon f(y) + P_\epsilon(fh)(y)\Big]\end{aligned}$$

Derivation: Substituting $\mathcal{L}_\epsilon$ into the CDC Operator (Part 2)

- **Substitute definition of $P_\epsilon g(y) = \frac{\mathbb{E}_X[w_\epsilon(y,X)g(X)]}{d_\epsilon(y)}$:**

$$\Gamma_\epsilon(f,h)(y) = \frac{1}{2\epsilon d_\epsilon(y)}\Big(f(y)h(y)d_\epsilon(y) - f(y)\mathbb{E}_X[w_\epsilon(y,X)h(X)] - h(y)\mathbb{E}_X[w_\epsilon(y,X)f(X)] + \mathbb{E}_X[w_\epsilon(y,X)f(X)h(X)]\Big)$$

- **Expand $d_\epsilon(y) = \mathbb{E}_X[w_\epsilon(y,X)]$ and consolidate into a single expectation:**

$$\Gamma_\epsilon(f,h)(y) = \frac{1}{2\epsilon d_\epsilon(y)}\mathbb{E}_X\Big[w_\epsilon(y,X)\big(f(y)h(y) - f(y)h(X) - h(y)f(X) + f(X)h(X)\big)\Big]$$

- **Factor the polynomial to obtain the marginal CDC:**

$$\Gamma_\epsilon(f,h)(y) = \frac{\mathbb{E}_X\big[w_\epsilon(y,X)\big(f(X)-f(y)\big)\big(h(X)-h(y)\big)\big]}{2\epsilon\mathbb{E}_X[w_\epsilon(y,X)]} \tag{5}$$

The Learning Problem: Marginal CDC Matching

- **Goal:** Train a neural network Γ_ϵ^θ to regress the marginal CDC operator $\Gamma_\epsilon(f, h)$ under smoothed density $p_Y = p * \mathcal{N}(0, \epsilon \boldsymbol{I})$.
- **Marginal Loss:**

$$\mathcal{L}_{marg}^{CDC}(\theta) := \mathbb{E}_Y \left[\left(\Gamma_\epsilon^\theta(Y) - \Gamma_\epsilon(f, h)(Y) \right)^2 \right] \tag{7}$$

- **Intractability:** The target $\Gamma_\epsilon(f, h)(Y)$ requires an evaluation of the expected value across the entire dataset for every single query point, preventing scalable mini-batch training.

Deriving Conditional CDC: Densities and Heat Kernel

- **Conditional Probability Density:** Given $X \sim p$ and noise-perturbed point $Y|X \sim \mathcal{N}(X, \epsilon \boldsymbol{I})$, the density is:

$$p_Y(y|x) = (2\pi\epsilon)^{-D/2} \exp\left(-\frac{\|x-y\|^2}{2\epsilon}\right)$$

- **Relation to Heat Kernel:** The Gaussian heat kernel $w_\epsilon(y, X)$ is strictly proportional to this conditional density:

$$w_\epsilon(y, X) = (2\pi\epsilon)^{D/2} p_Y(y|X)$$

- **Denominator Marginalisation:** The expectation of the conditional density over the prior $p(X)$ yields the marginal density $p_Y(y)$:

$$d_\epsilon(y) = (2\pi\epsilon)^{D/2} \mathbb{E}_X[p_Y(y|X)] = (2\pi\epsilon)^{D/2} p_Y(y)$$

Deriving Conditional CDC: Bayes' Rule Substitution

- **Tractable Conditional CDC Function:** Define a pairwise, constant-time target function:

$$\Gamma_\epsilon(f,h)(x,y) := \frac{1}{2\epsilon}(f(x)-f(y))(h(x)-h(y))$$

- **Substitute Density Relations into Marginal CDC (Eq. 5):**

$$\Gamma_\epsilon(f,h)(y) = \frac{(2\pi\epsilon)^{D/2}\mathbb{E}_X\left[p_Y(y|X)\Gamma_\epsilon(f,h)(X,y)\right]}{(2\pi\epsilon)^{D/2}p_Y(y)} = \frac{\mathbb{E}_X\left[p_Y(y|X)\Gamma_\epsilon(f,h)(X,y)\right]}{p_Y(y)} \tag{8}$$

- **Apply Bayes' Theorem:** Expand the expectation and substitute the posterior density $p_{X|Y}(x|y) = \frac{p_Y(y|x)p(x)}{p_Y(y)}$:

$$\Gamma_\epsilon(f,h)(y) = \int \Gamma_\epsilon(f,h)(x,y)\frac{p_Y(y|x)p(x)}{p_Y(y)}dx$$

$$= \int \Gamma_\epsilon(f,h)(x,y)p_{X|Y}(x|y)dx$$

- **Final Conditional Expectation:**

$$\Gamma_\epsilon(f,h)(y) = \mathbb{E}_{X|Y}\left[\Gamma_\epsilon(f,h)(X,y) \mid Y=y\right] \tag{9}$$

The Tractable Objective: Conditional CDC Matching

- **Conditional Loss Formulation:** Replace the marginal target with the constant-time conditional target, sampled jointly over the data X and the applied noise Y:

$$\mathcal{L}_{cond}^{CDC}(\theta) := \mathbb{E}_{X,Y|X}\left[\left(\Gamma_{\epsilon}^{\theta}(Y) - \Gamma_{\epsilon}(f,h)(X,Y)\right)^2\right] \tag{10}$$

- **Theorem 3.1 (Gradient Equivalence):** The conditional loss equals the marginal loss up to a constant C which is independent of the network parameters θ:

$$\mathcal{L}_{cond}^{CDC}(\theta) = \mathcal{L}_{marg}^{CDC}(\theta) + C \implies \nabla_{\theta}\mathcal{L}_{cond}^{CDC}(\theta) = \nabla_{\theta}\mathcal{L}_{marg}^{CDC}(\theta)$$

- **Computational Result:** Network optimisation bypasses explicit dataset-wide kernel construction and pairwise distance matrices, decomposing over independent mini-batches.

Proof of Theorem 3.1: Setup and Expansion

- **Variable Definition:** Let Z be the deterministic random variable representing the conditional CDC target, where $X \sim p$ and $Y \mid X \sim \mathcal{N}(X, \epsilon \boldsymbol{I})$:

$$Z := \Gamma_\epsilon(f, h)(X, Y)$$

- **Expectation Identity:** From Equation 9, the conditional expectation of Z given Y evaluates exactly to the marginal CDC:

$$\mathbb{E}_{X|Y}[Z \mid Y] = \Gamma_\epsilon(f, h)(Y)$$

- **Law of Total Expectation:** The unconditional expectation of the loss is rewritten as the expectation of the conditional expectation given Y:

$$\mathcal{L}^{CDC}_{cond}(\theta) = \mathbb{E}_{X,Y}\left[\left(\Gamma^\theta_\epsilon(Y) - Z\right)^2\right] = \mathbb{E}_Y\left[\mathbb{E}_{X|Y}\left[\left(\Gamma^\theta_\epsilon(Y) - Z\right)^2 \mid Y\right]\right]$$

Proof of Theorem 3.1: Bias-Variance Decomposition

- **Decomposition:** For a random variable Z and a constant c, the mean squared error decomposes as $\mathbb{E}[(c-Z)^2] = (c - \mathbb{E}[Z])^2 + \mathrm{Var}(Z)$. When conditioned on Y, the network prediction $\Gamma_\epsilon^\theta(Y)$ acts as a deterministic constant c:

$$\mathbb{E}_{X|Y}\left[\left(\Gamma_\epsilon^\theta(Y) - Z\right)^2 \mid Y\right] = \left(\Gamma_\epsilon^\theta(Y) - \mathbb{E}_{X|Y}[Z \mid Y]\right)^2 + \mathrm{Var}_{X|Y}(Z \mid Y)$$

- **Substitution and Final Derivation:**

$$\begin{aligned}
\mathcal{L}_{cond}^{CDC}(\theta) &= \mathbb{E}_Y\left[\left(\Gamma_\epsilon^\theta(Y) - \mathbb{E}_{X|Y}[Z \mid Y]\right)^2 + \mathrm{Var}_{X|Y}(Z \mid Y)\right] \\
&= \mathbb{E}_Y\left[\left(\Gamma_\epsilon^\theta(Y) - \Gamma_\epsilon(f,h)(Y)\right)^2\right] + \mathbb{E}_Y\left[\mathrm{Var}_{X|Y}(Z \mid Y)\right] \\
&= \mathcal{L}_{marg}^{CDC}(\theta) + \mathbb{E}_Y\left[\mathrm{Var}_{X|Y}(Z \mid Y)\right]
\end{aligned}$$

- **Conclusion:** The constant $C = \mathbb{E}_Y[\mathrm{Var}_{X|Y}(Z \mid Y)]$ depends only on the data distribution and noise, not on θ. Hence, the gradients are identical.

Derivation: Matrix-Valued Conditional CDC Target

- **Scalar Conditional CDC (Section 3.1):** For any scalar functions $f, h : \mathcal{M} \to \mathbb{R}$, the pairwise target is:
$$\Gamma_\epsilon(f,h)(X,Y) := \frac{1}{2\epsilon}\big(f(X) - f(Y)\big)\big(h(X) - h(Y)\big)$$
- **Ambient Coordinate Functions (Section 3.3):** To recover the full metric tensor $\mathbf{\Gamma}_\epsilon \in \mathbb{R}^{D \times D}$, apply the CDC to the ambient coordinate functions. Let f be the i-th coordinate X_i, and h be the j-th coordinate X_j:
$$\Gamma_\epsilon(x_i, x_j)(X,Y) = \frac{1}{2\epsilon}(X_i - Y_i)(X_j - Y_j)$$
- **Matrix Vectorisation:** The scalar output for coordinates i and j defines the (i,j)-th element of the target matrix $\boldsymbol{Z}$. Computing this for all $i, j \in \{1, \ldots, D\}$ yields the outer product:
$$\boldsymbol{Z}_{ij} = \frac{1}{2\epsilon}(X_i - Y_i)(X_j - Y_j) \implies \boldsymbol{Z} = \frac{1}{2\epsilon}(X - Y)(X - Y)^T$$
- **Matrix-Valued Loss (Eq. 11):** Substituting this aggregated target $\boldsymbol{Z}$ into the conditional loss yields:
$$\mathcal{L}_{cond}^{Riem} := \mathbb{E}_{X,Y|X}\left[\left\|\mathbf{\Gamma}_\epsilon^\theta(Y) - \frac{1}{2\epsilon}(X - Y)(X - Y)^T\right\|_F^2\right]$$